

Machine Learning

Introduction

Machine learning is a branch of artificial intelligence in which computers learn patterns from data and use those patterns to make predictions or decisions.

Supervised Learning

Supervised learning uses labeled training data. Common supervised learning tasks include classification and regression. Examples include spam detection and house price prediction.

Unsupervised Learning

Unsupervised learning works with data that does not have labeled answers. Clustering is a common example. It can be used to group customers with similar behavior.

Model Evaluation

Machine learning models should be evaluated using appropriate metrics. Classification may use accuracy, precision, recall, and F1-score, while regression may use mean absolute error or mean squared error.